

Coronary Heart Disease Mortality Trends in U.S. Adults Aged 35+

1. Introduction and background

The datasets from this analysis indicate heart disease has been the No.1 killer for the past few years while Coronary Heart Disease (CHD) is an essential component. This report analyzes the mortality rate and causes in the U.S., aiming to raise awareness of CHD and improve public health in a global context.

This topic was chosen for its significant public health impact on U.S. adults aged 35 and older. The study states the morbidity rate of people with CHD in different age groups. Analyzing CHD mortality trends assesses the effects of environment changes and health disparities on outcomes.

2. Objectives and goals

The primary objective of this project is to analyze trends in mortality related to cardiovascular disease, particularly coronary heart disease (CHD), among adults 35 years of age and older in the United States. We target to identify key factors contributing to mortality from coronary heart disease and explore how advances in healthcare and lifestyle changes have influenced these trends over time.

Through this project, we aim to gain insights which can benefit public health stakeholders, policymakers, and healthcare professionals in the following aspects.

- (1) To increase awareness of the long-standing risk factors for coronary heart disease (CHD). Persistent risk factors such as high blood pressure, obesity, smoking, poor diet and physical inactivity remain the leading causes of death from coronary heart disease (Jung & Yang, 2021). The project will analyze how these factors affect mortality, helping refine public health campaigns to address the most harmful risks and highlight emerging problems.

- (2) To identify people at high risk of cardiovascular disease for more targeted health interventions. Our analysis will identify high-risk populations or regions with higher rates of cardiovascular disease mortality. The findings will guide targeted health interventions, such as screening or education programs for at-risk populations. Data-driven strategies can help healthcare providers prioritize resources to reduce coronary heart disease mortality through focused efforts.

3. Visualization Plan

To comprehensively analyze the various influencing factors and trends of cardiovascular disease, we have designed six charts, including a line chart, bar chart, map, bar chart, scatter plot, and table, to explore the prevalence of CHD from multiple perspectives. By addressing both macro and micro levels, we conducted a detailed analysis of overall disease trends and specific hypothesized risk factors. The line chart shows the trend in heart disease rates from 1999 to 2017, and the table lists the top ten causes of death during the same period. The bar chart presents coronary heart disease rates by age group, while the map displays cardiovascular disease mortality rates by state. The bar chart compares the relationship between different blood pressure levels and CHD, and the scatter plot or table analyzes the impact of living environment (urban vs. rural) on cardiovascular disease. Together, these charts form a multidimensional analysis framework, providing a comprehensive view of cardiovascular disease trends and influencing factors.

4. Conclusion

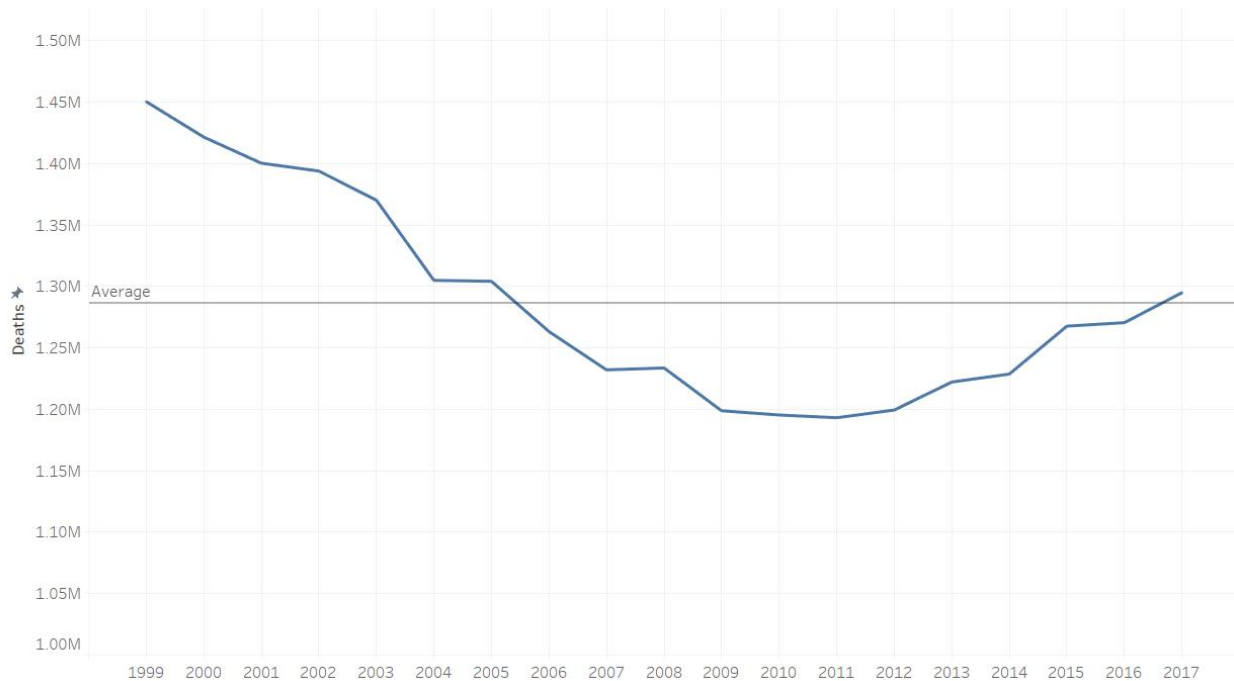
The focus on coronary heart disease is not only due to its high prevalence and status as a leading cause of death but also due to the broad range of modifiable factors that, if addressed effectively, can significantly reduce its impact. In the following analysis, several key factors have been identified that increase the risk of developing CHD including high blood pressure, poor air quality, aging, and the rising prevalence of obesity. Understanding the factors that contribute to CHD is critical for developing effective prevention and management strategies. By addressing these factors, there is a promising potential to not only improve individual health outcomes but also reduce the overall healthcare burden posed by coronary heart disease.

Attachment:

1. Datasets

1. Line Chart: Trend of Heart Disease from 1999 to 2017

Trend of Heart Disease From 1999 to 2017



Line chart: The chart analyzes total heart disease deaths year over year using the dataset from **National Center for Health Statistics**. This dataset covers 52 states of the United States including 10,868 deaths caused by heart disease from 1999 to 2017. The data was collected by the **National Center for Health Statistics (NCHS)** which is also a branch of Disease Control and Prevention (CDC). The project was funded by the **Centers for Disease Control and Prevention (CDC)**. The dataset was created to analyze the leading causes of death in the United States and help the public to get a better understanding of mortality trends and identify major public health concerns.

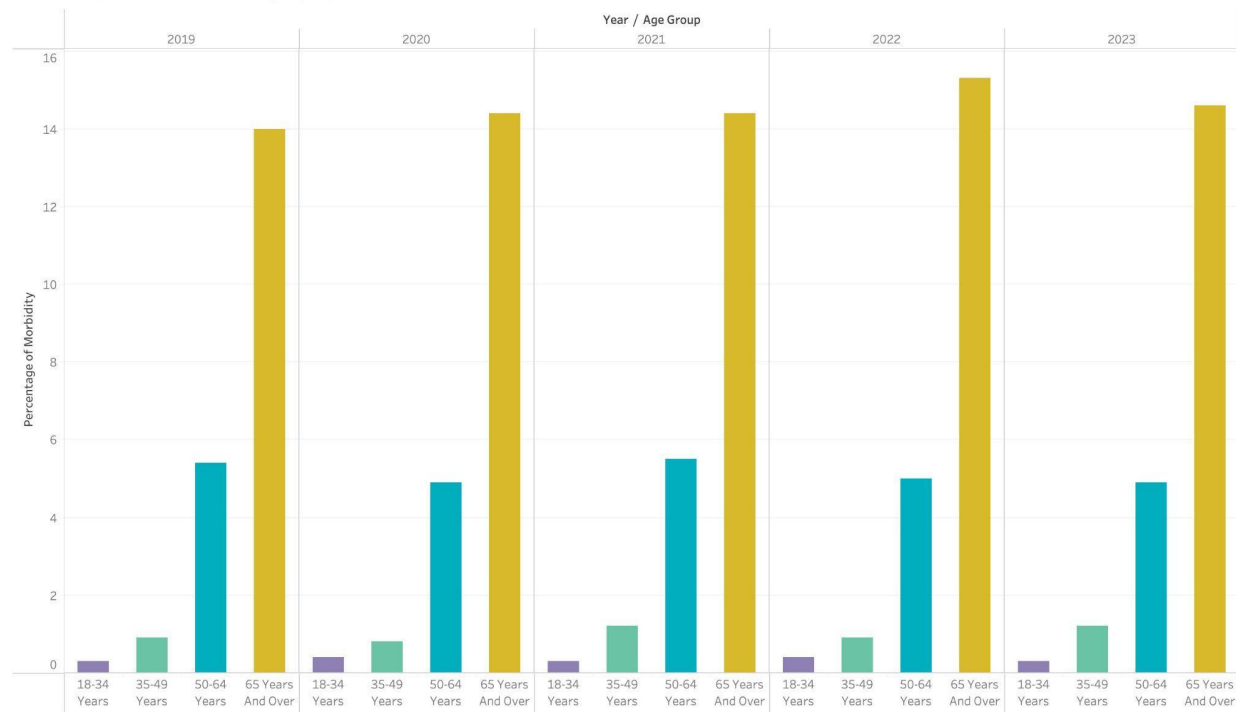
(Insight: The graph shows a clear decline in deaths from heart disease from 1999 until around 2011. This could indicate the impact of improved cardiovascular treatments, better public awareness about heart health, and possibly changes in lifestyle and diet that have become more prevalent over the years.

After 2011, there was a noticeable plateau and then a slight increase in deaths. The rise may suggest new challenges in the public health domain, such as the obesity epidemic, diabetes, and other lifestyle diseases that might contribute to worsening heart health in the general population.

The recent trends of stabilization and a slight increase in heart disease deaths highlight the need for renewed focus in both healthcare and policy domains to address the root causes. It suggests that while earlier efforts had significant positive effects, heart disease remains a critical area requiring continued resource allocation and innovation in treatment and prevention strategies.)

2. Bar Chart: Percentage of CHD Morbidity by Age

Percentage of CHD Morbidity by Age

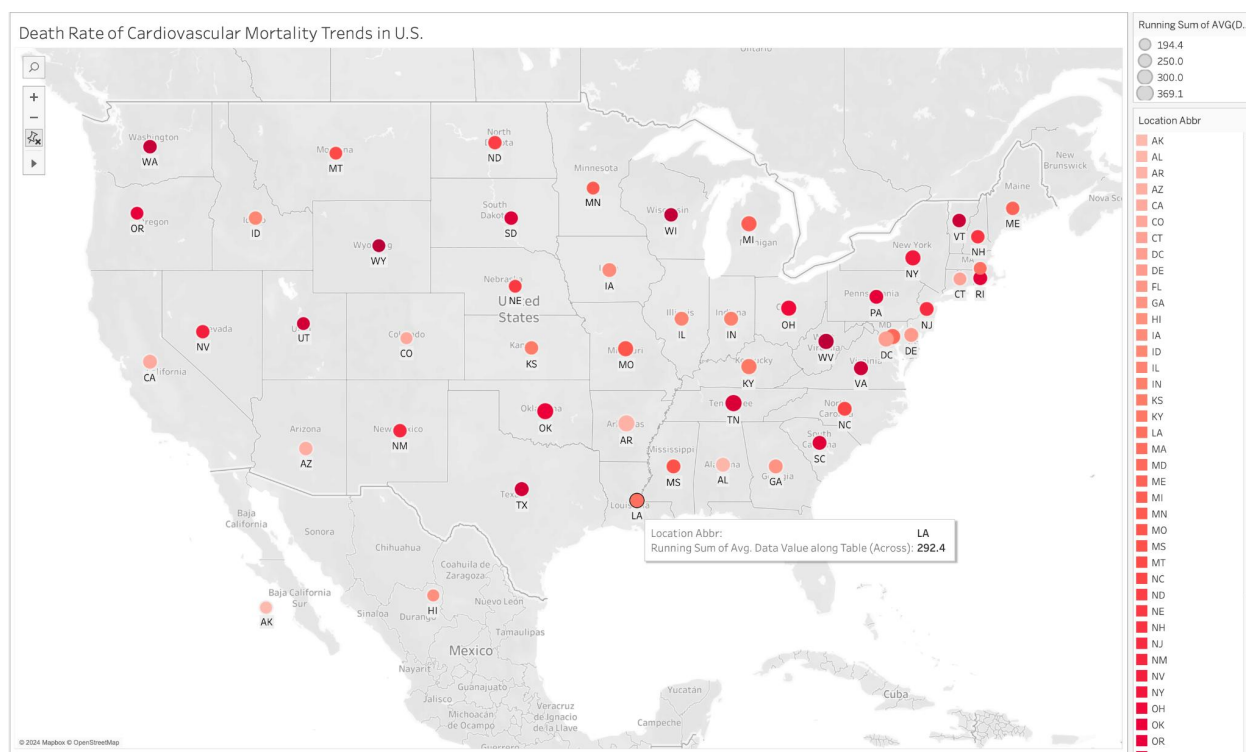


Bar chart: Percentage of morbidity refers to the percentage of people with coronary heart disease in each age group compared to the total population of that age group. The age groups are categorized as 18-34, 35-49, 50-64, and 65 and older. The data covers a five-year period from 2019 to 2023.

(Insight: CHD morbidity is significantly higher in older age groups (65-74 and 75+ years old), with these groups consistently showing the highest percentages across all observed years. This trend is consistent with the general medical understanding that the risk of coronary heart disease increases with age.

For each age group, the percentage of CHD morbidity appears relatively stable from 2019 to 2023, suggesting that there hasn't been a dramatic change in the incidence rates within these years. This could imply that, during this period, there were no significant public health interventions or changes in healthcare practices that drastically altered the morbidity rates of CHD. The pronounced prevalence of CHD in the elderly underscores the importance of targeted preventive measures in these age groups.)

3. Map Chart: Death Rate of Cardiovascular Mortality Trends in U.S.



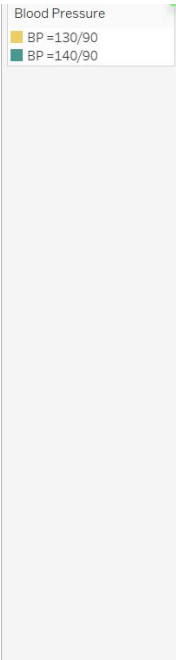
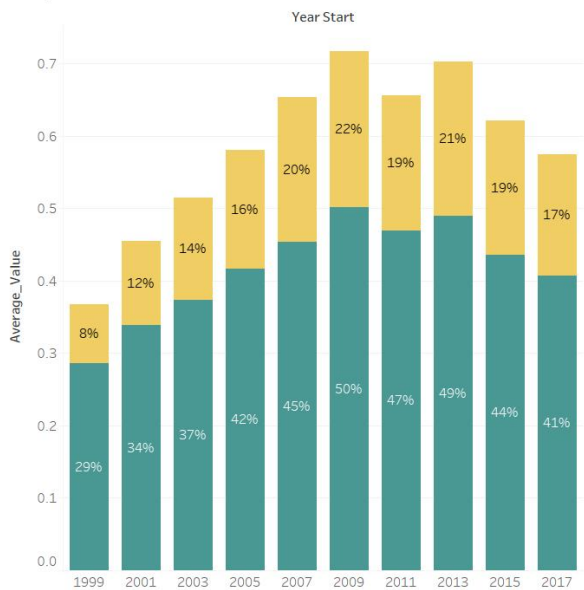
The dataset on coronary heart disease and stroke mortality trends among U.S. adults aged 35 and older was sourced from the National Vital Statistics System (NVSS), likely managed by the CDC. It was collected to

monitor and analyze cardiovascular mortality trends across U.S. counties, providing critical insights for public health research and policy. Spanning from 1999 to 2018, the dataset includes 288,512 entries with 20 variables, covering important details such as geographic location, year, mortality rates, and disease categories. It offers a comprehensive view of mortality trends at the county level, allowing for detailed analysis and visualizations at the county, state, and national levels. This dataset is essential for understanding the long-term patterns of heart disease and stroke mortality and has likely informed several public health publications.

(Insight: The relatively uniform size of the circles across the map suggests that cardiovascular disease affects a broad demographic and is not confined to specific geographic or socio-economic boundaries within the U.S. This indicates a nationwide impact that transcends regional lifestyle and healthcare differences.)

4. Bar Chart: Annual Coronary Heart Disease Rates by Blood Pressure Levels (130/90 vs 140/90): A Bar Chart Comparison

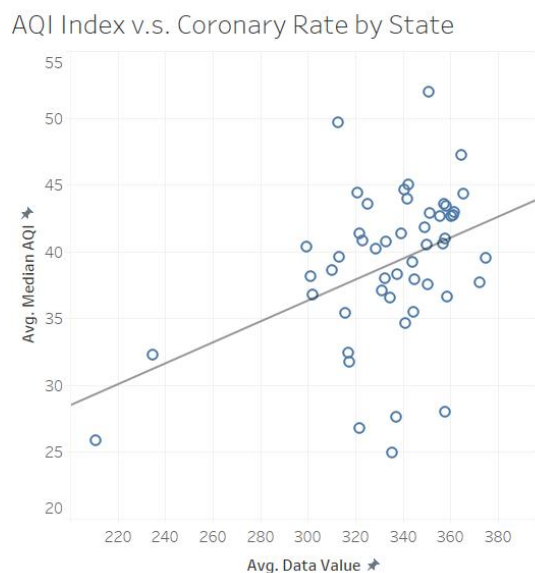
Annual Coronary Heart Disease Rates by Blood Pressure Levels (130/90 vs 140/90): A Bar Chart Comparison



This bar chart uses data from the **Centers for Disease Control and Prevention (CDC)**, titled "Rates and Trends in Coronary Heart Disease and Stroke Mortality in U.S. Counties, 1999-2018." The dataset was created to analyze coronary heart disease and stroke mortality trends at the county level in the U.S., aimed at informing public health policy decisions. The data spans from 1999 to 2018, covering over 3,000 counties and providing mortality rates based on age, gender, race, and geographic location. The chart displays two variables: **Year** and **Average Value** (average disease rate), comparing two blood pressure levels: **130/90** and **140/90**.

(Insight: Higher blood pressure (140/90) is consistently associated with higher rates of coronary heart disease, underscoring the direct impact of elevated blood pressure levels on heart health. The data highlight the importance of managing blood pressure within safer thresholds to potentially reduce the incidence of coronary heart disease.)

5. Scatter Plot: AQI Index v.s. Coronary Rate by State



The scatter plot "AQI Index vs. Coronary Rate by State" shows the relationship between Air Quality Index (AQI) levels and coronary morbidity rates across different U.S. states for the year

2015. This visual analysis seeks to explore how variations in air quality may correlate with the incidence of coronary diseases, suggesting potential environmental impacts on public health.

(Insight: The trend line in the scatter plot shows a positive relationship, indicating that as AQI increases, implying poorer air quality, there tends to be a higher rate of coronary morbidity. This trend suggests that states with worse air quality might experience higher rates of heat-related illnesses. This visual analysis seeks to explore how variations in air quality may correlate with the incidence of coronary diseases, suggesting potential environmental impacts on public health.)

6. Table Chart: Top 10 Leading Death Causes From 1999 to 2017

Top 10 Leading Death Causes From 1999 to 2017

Cause Name	
Heart disease	24,445,280
Cancer	21,687,288
Stroke	5,453,046
CLRD	5,189,854
Unintentional injuries	4,695,640
Alzheimer's disease	2,989,632
Diabetes	2,799,886
Influenza and pneu..	2,189,282
Kidney disease	1,717,226
Suicide	1,394,032

Table Chart: Table 6 displays top 10 leading causes of deaths in the United states from 1999 to 2017 covering 10,868 cases. This chart is using the same data source as Table 1 which includes cause name and total deaths to analyze whether heart disease is one of the major leading causes for death. The dataset used comes from the **National Center for Health Statistics (NCHS)** titled *Leading Causes of Death: United States*, which is available on Data.gov.

(Insight: Heart disease is one of the leading causes in the U.S., and coronary artery disease is the most common type of heart disease, emphasizing the critical importance of addressing cardiovascular health.)

Reference

1. Lake County Illinois GIS. (2015). National obesity by state. Data.gov.
<https://catalog.data.gov/dataset/national-obesity-by-state-d765a>
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3. U.S. Department of Health and Human Services. (n.d.). National Health and Nutrition Examination Survey (NHANES): National cardiovascular disease surveillance. Data.gov.
<https://catalog.data.gov/dataset/national-health-and-nutrition-examination-survey-nhanes-national-cardiovascular-disease-su>
4. Centers for Disease Control and Prevention. (n.d.). Rates and trends in coronary heart disease and stroke mortality in U.S. counties, 1999-2018. <https://data.cdc.gov/Heart-Disease-Stroke-Prevention/Rates-and-Trends-in-Coronary-Heart-Disease-and-Str/9cr5-2tt7/about>
5. National Center for Health Statistics. (2024). Percentage of angina for adults aged 18 and over, United States, 2019–2023: National Health Interview Survey. Retrieved September 22, 2024, from https://wwwn.cdc.gov/NHISDataQueryTool/SHS_adult/index.html
6. Jung, H. G., & Yang, Y. K. (2021). Factors influencing health behavior practice in patients with coronary artery diseases. *Health and Quality of Life Outcomes*, 19(3).
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7. U.S. Environmental Protection Agency. (n.d.). Download files.
https://aqs.epa.gov/aqsweb/airdata/download_files.html#Annual
8. National Center for Health Statistics. (2022, April 21). Leading causes of death: United States. Data.gov. <https://catalog.data.gov/dataset/nchs-leading-causes-of-death-united-states>

3. Teams Contribution

Team Member	Assigned Task	Chart Title	APA References to Find
Yahuan Peng	Background and motivation: Write the background section of the project, discussing mortality trends of cardiovascular disease among adults aged 35 and older in the U.S. and the research motivation.	1. Line Chart: Trend of Heart Disease from 1999 to 2017. 6. Top 10 Leading Death Causes From 1999 to 2017.	Find literature on overall cardiovascular disease trends and their main causes.
Yao Liu	Objective and significance: Clarify project objectives, analyzing how to improve public health.	2. Percentage of CHD Morbidity by Age.	Find literature on cardiovascular disease incidence across different age groups.
Huangkun Chen	Dataset description: Provide a detailed description of the dataset and its source of variables.	3. Cardiovascular Disease Mortality Rates by State Map.	Find literature on geographic location and the distribution of cardiovascular disease.
Yingtong Wang	Visualization design and divide team tasks: Describe how the six charts help achieve the project goals.	4. Annual Coronary Heart Disease Rates by Blood Pressure Levels (130/90 vs 140/90).	Find literature on the relationship between blood pressure and cardiovascular disease.
Xiaoting Lu	Result analysis and discussion: Analyze the results of the charts and discuss their implications for the research objectives.	5. Scatter Plot Showing the Impact of Living Environment (AQI) on Cardiovascular Disease.	Find literature on the impact of living environment on health.